

YINING, China

WMO: 514310

Lat: 43.9554N

Lon: 81.3363E

Elev: 664

StdP: 93.60

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-18.2	-23.6	0.5	-20.7	-20.6	0.6	-17.7	5.5	-4.5	4.4	-3.4	0.7	70	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	34.0	19.1	32.6	18.6	31.3	18.1	20.7	30.2	20.0	29.6	19.3	28.6	1.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	13.8	24.4	16.8	13.0	23.6	16.0	12.3	23.1	63.3	30.2	60.4	29.8	57.9	28.5	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.2	5.0	4.1	-23.3	36.6	3.5	0.9	-25.8	37.3	-27.8	37.8	-29.8	38.3	-32.3	39.0		
			-23.5	22.2	3.4	1.3	-26.0	23.1	-28.0	23.8	-29.9	24.5	-32.4	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-7.3	-3.6	5.7	13.3	18.3	22.4	24.0	22.8	18.0	11.0	3.6	-4.2	(6)
(7)		DBStd	11.57	5.06	4.89	5.39	4.39	3.50	2.77	2.49	2.78	3.53	4.11	4.60	4.75	(7)
(8)		HDD10.0	1765	537	381	153	20	1	0	0	1	36	196	441		(8)
(9)		HDD18.3	3425	795	614	392	155	44	4	1	48	227	442	699		(9)
(10)		CDD10.0	1912	0	0	19	121	256	371	434	398	241	68	4	0	(10)
(11)		CDD18.3	530	0	0	0	6	42	124	177	142	38	1	0	0	(11)
(12)		CDH23.3	5808	0	0	5	127	493	1221	1827	1582	519	33	0	0	(12)
(13)		CDH26.7	2295	0	0	0	27	138	462	812	692	161	3	0	0	(13)
(14)	Wind	WSAvg	1.7	1.2	1.4	2.0	2.3	2.1	2.0	1.9	1.8	1.6	1.4	1.4	1.3	(14)
(15)	Precipitation	PrecAvg	272	18	19	21	29	30	28	25	15	15	25	31	24	(15)
(16)		PrecMax	470	94	65	86	99	67	71	80	46	47	86	115	68	(16)
(17)		PrecMin	138	1	1	0	0	0	1	3	1	0	3	2	0	(17)
(18)		PrecStd	82	14	12	16	19	16	18	17	11	10	19	23	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	10.0	23.3	29.6	32.1	34.1	36.4	35.6	32.5	26.7	19.0	8.9	(19)
(20)			MCWB	1.7	4.9	10.9	14.4	16.7	18.4	20.3	19.4	18.1	14.3	9.6	3.6	(20)
(21)		2%	DB	2.9	6.9	19.8	26.7	30.0	32.4	34.5	33.7	30.3	23.8	15.7	5.7	(21)
(22)			MCWB	0.0	2.7	9.9	13.1	16.1	18.2	19.6	19.0	16.9	13.1	8.2	1.7	(22)
(23)		5%	DB	1.3	4.7	17.3	24.4	28.1	31.0	32.8	32.3	28.6	21.2	12.6	3.8	(23)
(24)			MCWB	-0.8	1.4	9.0	12.4	15.6	18.1	19.0	18.6	16.2	11.9	7.1	0.7	(24)
(25)		10%	DB	0.0	2.9	14.6	22.0	26.2	29.6	31.1	30.7	26.6	18.8	10.1	1.8	(25)
(26)			MCWB	-1.6	0.3	7.9	11.6	14.9	17.5	18.5	18.1	15.4	10.9	5.9	-0.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	5.1	12.1	16.2	19.0	21.1	22.4	21.5	19.2	15.0	10.9	4.2	(27)
(28)			MCDWB	4.6	9.2	20.7	25.9	28.4	29.9	31.9	30.7	30.1	25.0	16.0	7.6	(28)
(29)		2%	WB	0.7	3.1	10.6	14.4	17.5	19.9	21.1	20.5	17.8	13.8	9.0	2.5	(29)
(30)			MCDWB	2.3	6.1	18.5	23.9	26.7	28.8	30.8	30.0	28.1	22.2	14.3	5.2	(30)
(31)		5%	WB	-0.4	1.8	9.4	13.2	16.6	19.1	20.2	19.7	16.8	12.7	7.6	0.9	(31)
(32)			MCDWB	1.0	4.2	16.4	21.7	25.5	28.1	29.9	29.5	26.8	19.7	12.0	3.2	(32)
(33)		10%	WB	-1.6	0.6	8.2	12.4	15.7	18.4	19.5	18.9	15.9	11.6	6.2	-0.1	(33)
(34)			MCDWB	-0.1	2.6	13.7	20.0	24.0	27.0	28.7	28.5	25.1	17.5	9.9	1.5	(34)
(35)	Mean Daily Temperature Range	MDBR	10.5	9.3	10.5	12.3	12.4	12.0	12.5	13.7	14.4	13.4	9.9	9.3	(35)	
(36)		5% DB	MCDBR	11.0	10.8	14.7	17.0	15.4	14.4	15.3	16.3	17.5	17.3	14.8	11.6	(36)
(37)			MCWBR	8.2	7.1	7.3	7.0	5.8	5.1	5.2	6.2	7.2	7.7	7.9	7.9	(37)
(38)		5% WB	MCDBR	8.6	9.1	13.5	13.9	12.9	12.4	13.3	14.5	16.3	15.0	13.3	9.8	(38)
(39)			MCWBR	6.5	6.1	6.8	6.0	5.2	4.8	5.2	5.9	7.1	6.7	7.3	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.371	0.432	0.472	0.443	0.439	0.426	0.404	0.391	0.384	0.361	0.321	(40)	
(41)		taud	2.174	2.074	1.973	1.922	2.068	2.144	2.208	2.256	2.230	2.153	2.139	2.178	(41)	
(42)		Ebn at Noon	772	791	788	791	831	835	842	848	827	768	713	730	(42)	
(43)		Edh at Noon	94	125	158	181	161	150	139	128	121	114	97	85	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.14	3.08	4.10	5.15	6.28	6.45	6.44	5.97	4.88	3.23	2.01	1.70	(44)	
(45)		RadStd	0.09	0.15	0.27	0.40	0.36	0.26	0.36	0.28	0.28	0.29	0.16	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.55	+2.18	N/A	+0.47	-0.54	-0.78	-142	-165	+117	+64
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	-0.42	N/A	N/A	N/A	+94	N/A

Nomenclature: See separate page